

MUDITHA PRIYASAD

Mechanical Design, Automation & R&D Engineer

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PROFESSIONAL SUMMARY

Hands-on Mechanical Engineer and consultant with practical experience in machine design, factory automation, PLC-based controls, pneumatic systems, troubleshooting and rapid prototyping. Builds production machines from concept through commissioning, testing and operator-focused improvement, connecting mechanical design with automation and manufacturing needs.

Highlights: 15+ practical projects | 4 industrial environments | 5+ industrial training programs | Multi skilled

CORE SKILLS

Machine Design: Jigs, fixtures, feeders, frames, mechanisms, design concepts, SolidWorks/3D CAD, 2D/fabrication drawings, DfMA/DfAM, material and process awareness

Automation & Controls: PLC programming (Siemens S7-200, Delta, Xinje, Mitsubishi), sensors, actuators, pneumatics, electrical wiring, control panels, commissioning support

Production Improvement: Industrial troubleshooting, root-cause analysis, mechanical evaluations, preventive-maintenance awareness, TPM, 5S, Kaizen, reliability improvement

Prototyping: 3D printing, CNC laser systems, G-code workflows, workshop fabrication, fit/function testing, iterative modification and low-cost validation

Digital Tools: Raspberry Pi, Arduino, Python, C++, Microsoft Office, dashboard/reporting work, AI-assisted engineering and workflow tools

PROFESSIONAL EXPERIENCE

Consultant Engineer — Mechanical Design & Automation

MARQO Industries Pvt Ltd | 2024 – Present

- Increased fan-base production output by ~400%, raising throughput from 1 to 6 units per minute, by developing and commissioning a guided filling and assembly machine for oil application, nut tightening and part alignment.
- Increased B22 eyelet punching speed by 3x and reduced operator effort by designing an ergonomic automated puncher with optimized mechanical action.
- Improved fan motor assembly efficiency and reduced operator strain by creating a guided fixture with hands-free oil application and assisted fastener tightening.
- Supported commissioning, setup, optimization and troubleshooting of plug-top packing and spot-welding equipment, including design concepts, prototype parts and fabrication guidance for production support tools.
- Established the engineering and automation department from the ground up — building workshop guides, department software, standardized tools and core processes from scratch with no prior team or systems in place.

Founder / Designer

MTone Designers | 2021 – Present

- Design machine and product concepts using 3D printing, CNC laser engraving, wood-burning artwork, custom fixtures and small-scale machine development.
- Build and test custom 3D printer and CNC laser concepts across frame design, motion systems, electronics integration.

Engineering Intern

MAS Active Shadowline | 2024 (9 months)

- Contributed to TPM, 5S and Kaizen initiatives by developing machine standards, checklists, SOPs, dashboards and handover documents that strengthened maintenance and engineering workflows.
- Temporarily supported Sustainability Executive responsibilities during a staffing transition by tracking electricity bills, water bills, waste records and KPI reports.
- Prepared CAD layouts, tool/equipment lists, budget inputs, training material, visual-management labels and production-support documentation.

Automation Engineering Intern

Effective Engineering & Automation Pvt Ltd | 2023 (6 months)

- Built hands-on capability across industrial automation, machine troubleshooting, electrical wiring, PLC programming, equipment testing and documentation.
- Worked on machine and test-equipment projects including a 3D printer, line-follower robot, screen printer, concrete mixers and glove-making robot support tasks.

Autonomeering Engineering Intern

MAS Kreedaa Vaanavil | 2019 (3 months)

- Applied pneumatic circuit design, CAD, PLC ladder logic, Arduino programming, power tools and lathe-machine operation across hands-on workshop tasks.
- Supported boiler monitoring, bonding-tape machine development and fabric-picking mechanism research.

KEY PROJECTS

- **Fan Base Filling Machine** — designed a guided filling, oil-application and assembly rig integrating part alignment and nut tightening for high-throughput production.
- **Custom 3D Printer** — built and tested a working machine concept covering frame design, motion control, electronics integration and practical prototyping support.
- **1W LED Bulb Assembly Machine** — developed an automated assembly concept to reduce repetitive handling, guide parts correctly and improve production consistency.
- **B22 Eyelet Puncher** — engineered an ergonomic automated punching mechanism with optimized mechanical action for repetitive eyelet work.
- **PCB Drilling Machine** — CNC-controlled rig built for NDM Electrical Pvt Ltd, drilling 18 PCB pieces in one setup and improving alignment, accuracy, repeatability and finish quality while reducing manual work.
- **Bulb Assembly Machines** — set up and troubleshoot two types of bulb-assembling machines for NDM Electrical Pvt Ltd, supporting reliable production operation.

Additional project details available at muditha96.github.io.

EDUCATION

BSc (Hons) in Mechanical Engineering

University of Jaffna — Faculty of Engineering | October 2017 – February 2024

CERTIFICATIONS

- SOLIDWORKS CSWA — CAD Design
- SOLIDWORKS CSWA-AM — Additive Manufacturing
- PLC Programming Course — Epic Engineering, 2024
- PLC Training — University of Moratuwa, 2025
- Maintenance of Industrial Plant & Machinery — NERDC
- Industrial Motor Operation & Control System — NERDC
- Advanced Injection Molding — IDB

ACHIEVEMENTS & ACTIVITIES

- Bronze Award – Best Display of Engineering Projects, Techno Sri Lanka 2023
- Participant – AIR TATTOO Exhibition 2024
- Participant – SLT Robot Battles 2019
- Organizer – UOJ BATTLEGROUNDS 2020
- Languages – English, Sinhala

Scan for portfolio

